Arnav Makarand Yadnopavit

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.58
Podar International School, Chinchwad	Class XII (CBSE)	2024	96.00%
Podar International School, Chinchwad	Class X (CBSE)	2022	98.00%

SKILLS

Programming Languages: C, C++, Python, Verilog, SystemVerilog, MATLAB, RISC-V Assembly, Shell, SQL

EDA & Design Tools : Vivado, LTSpice, KiCad, Cadence Virtuoso

Python Libraries: PyTorch, NumPy, Matplotlib, Pandas, Scikit-learn, SciPy

Other Tools: Git, LaTeX, Arduino IDE, PlatformIO

OS: Windows, Ubuntu

RELEVANT COURSES

Computer Science Courses: Computer Architecture, Data Structures and Applications, Introduction to Programming

Math Courses: Mathematics for EE, Matrix Theory, Probability, Differential Equations and Transforms, Random Processes

Core Courses: Digital Systems, Machine Learning, Analog Electronics, Digital Signal Processing, Control Systems

PROJECTS

NeuraCore -- Custom CNN Inference Accelerator Development board using Zynq-7000 SoC *(Ongoing)*

- Implemented **Eyeriss**-based CNN accelerator RTL with **row-stationary dataflow** and configurable **systolic array** in **SystemVerilog**, synthesized on **Xilinx Zynq-7000 SoC** using **Vivado**
- Verified **PE array** and dataflow control logic correctness by authoring **SystemVerilog testbenches** and meeting **timing closure** under Vivado constraints
- Deployed a quantized **CNN OCR** model onto **Zynq-7000 PL** by designing architecture, training, and performing **fixed-point vs floating-point analysis** to determine optimal **quantization** scheme

RISC-V Processor on FPGA (RV64IM)

- Designed a **64-bit 5-stage pipelined RV64IM** processor in **Verilog** with full **data forwarding** and **load-use hazard detection**, achieving **30 MHz** on **Artix-7 100T FPGA** via **Vivado**
- Validated functional correctness across bubble sort, GCD, and memory stress testbenches, utilizing **~14k LUTs**, **~13K FFs**, **54 DSP slices**, and **32 BRAMs** on Arty A7
- Implemented **multicycle divider FSM** using Xilinx **div_gen IP** to support the **M-extension** (integer multiply/divide)

Embedded Systems

- Developed a **FreeRTOS application on Raspberry Pi Pico** featuring task scheduling, queues, and interrupt-driven peripherals.
- Built an **Arduino Scientific Calculator** implementing CORDIC, RK4, custom expression parsing, and LCD interfacing.
- Designed a **Multiplexed Arduino Digital Clock** using Timer0 interrupts, seven-segment multiplexing, and button debouncing.

Machine Learning

- Implemented **regression models** with polynomial and Gaussian basis functions, SGD, MAP estimation, and bias-variance analysis.
- Developed **classification algorithms** including least-squares, Fisher's LDA, logistic regression, and perceptron-based classifiers.
- Implemented **Q-learning** for MDP policy optimization and a **feedforward neural network** from scratch for MNIST digit classification.

QR Eigenvalue Solver (Numerical Linear Algebra)

- Computed eigenvalues of general **n×n matrices** in **C** by implementing **QR Algorithm** with **Householder transformations** and **Wilkinson Shift**, achieving **quadratic convergence** with **$O(n^3)$** complexity
- Supported **complex eigenvalues** by extracting **2×2 Hessenberg blocks** along the diagonal, validated results against **NumPy** eigenvalue outputs across multiple test matrices

ACHIEVEMENTS

Runner-Up, FIFS Sports Data Gameathon *(Dream11 & FIFS, Waves Summit 2025)*

Represented IIT Hyderabad as team *ML Mavericks*; built a machine learning model to maximize fantasy cricket points across the ICC Champions Trophy 2025, competing against top engineering institutes nationally

Bronze, Capture The Flag -- Milan Inter-Hostel Cybersecurity Competition*(IIT Hyderabad)*

Secured team bronze in a CTF cybersecurity challenge competing across hostels

POSITIONS OF RESPONSIBILITY

Senior Core Member, Elektronica -- Electronics & tech club of IITH

PRN Coordinator, TEDxIITHyderabad -- Managed public relations and outreach for TEDx events; previously served as Assistant Coordinator

Teaching Assistant -- Matrix Theory -- Assisted course instruction for undergraduate Matrix Theory at IITH